Theory of Computation

Lecture 12

Computation and Its Limitations!

$$A_{\mathrm{TM}} = \{\langle M, w\rangle : M \text{ accepts } w\}$$

$$\mathrm{ALL}_{\mathrm{CFL}} = \{\langle G\rangle : \mathscr{L}(G) = \Sigma^*\}$$

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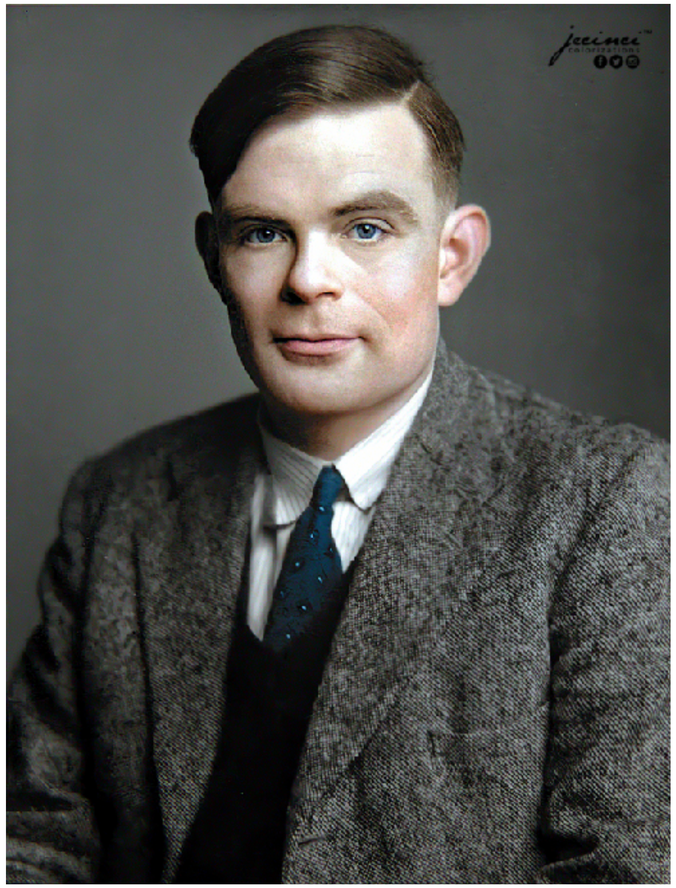

Alan Turing

On Computable Numbers, with an Application to the Entscheidungsproblem

Every DFA has a corresponding (right-linear grammar)

Every CFL has a corresponding (halting) TM

Which of the following are undecidable?

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Every CFL has a corresponding (halting) TM

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$$\mathrm{A}_{\mathrm{CFL}} = \{\langle G, w\rangle : w \in \mathscr{L}(G)\}$$

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Which of the following are undecidable?

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Which way do we reduce?

Turing Reduction (Oracle Reduction)

If problem A reduces to B, then we denote it by $A \leq_T B$.

That is, given an oracle for B, we can solve A.

If B is decidable then so is A.

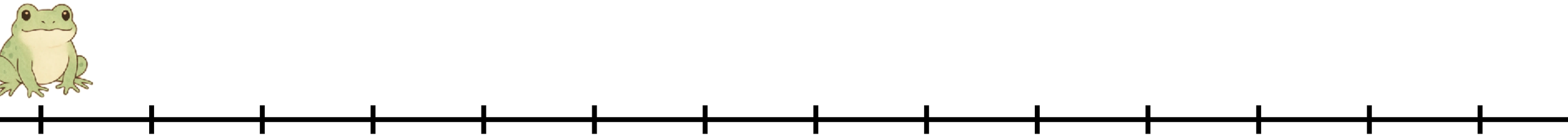
If A is undecidable then so is B.

$$\mathrm{ALL}_{\mathrm{DFA}} \leq_T \mathrm{ALL}_{\mathrm{CFL}} \leq_T \mathrm{ALL}_{\mathrm{TM}}$$

Hunting a Frog

Frog picks a strategy $k \in \mathbb{N}^{\geq 1}$

At time t, frog jumpts to kt



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$k = 3$

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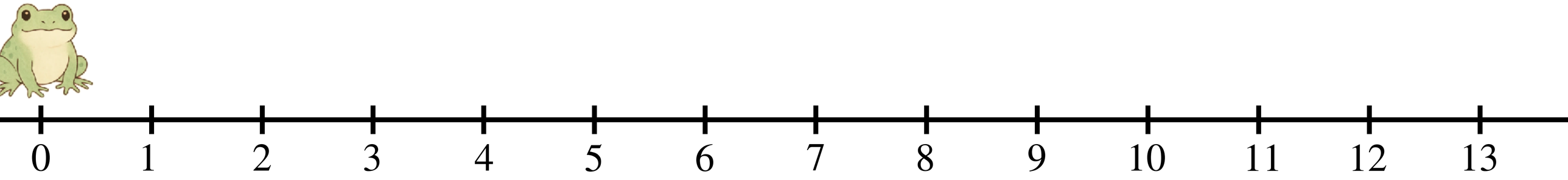


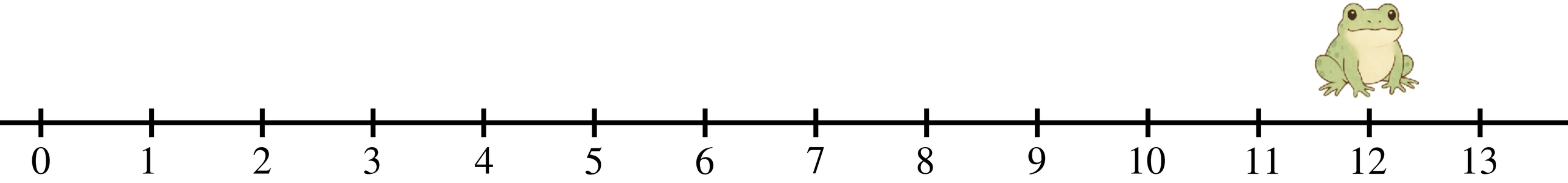
Hunting a Frog

$k = 3$

Frog picks a strategy $k \in \mathbb{N}^{\geq 1}$

At time t, frog jumpts to kt

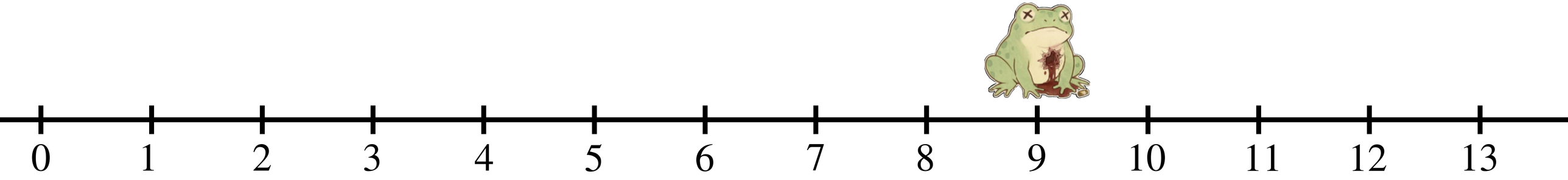
Prove that there is a fixed hunting strategy
that works regardless of what the frog does.



Strategy of Squares

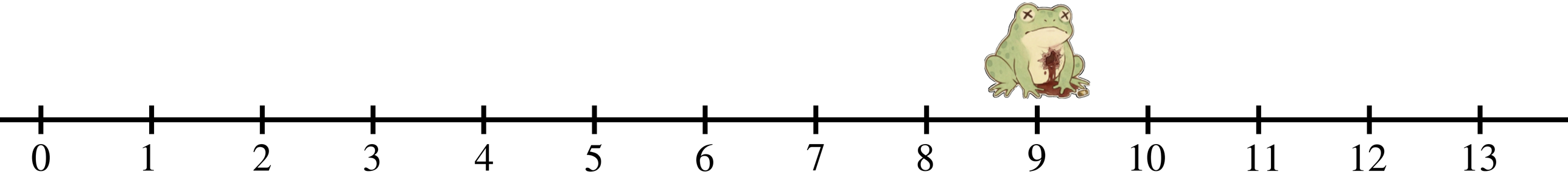
At time t, the hunter shoots at t^2

If the frog chose k, it will be shot at time k at position k^2



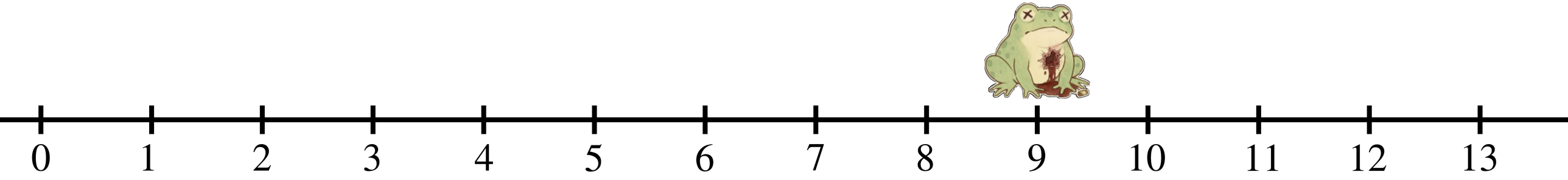
Strategy of Squares

Even after knowing the strategy of squares,
why can the frog not outsmart the hunter?



Strategy of Squares

Suppose the frog can pick any polynomial $p : \mathbb{Z} \to \mathbb{Z}$, then?



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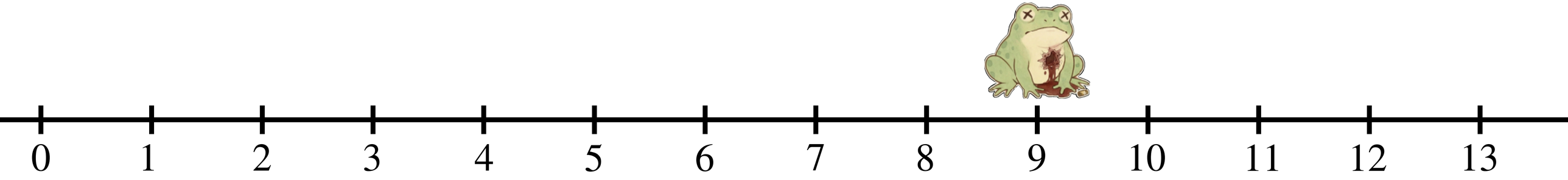
Suppose the frog can pick any polynomial $p : \mathbb{Z} \to \mathbb{Z}$, then?

Can AI save the frog?



MOLLY TAFT SCIENCE FEB 18, 2026 10:17 AM

Big Tech Says Generative AI Will Save the Planet. It Doesn't Offer Much Proof

A new report finds that of 154 specific claims about how AI will benefit the climate, just a quarter cited academic research. A third included no evidence at all.

Despite its steep environmental costs, AI might also help save the planet

Published: January 21, 2026 1.35pm GMT

BEZOS EARTH FUND

WHO WE ARE OUR APPROACH

Harnessing AI to Help Save the Planet

INSIGHT OCTOBER 16, 2025

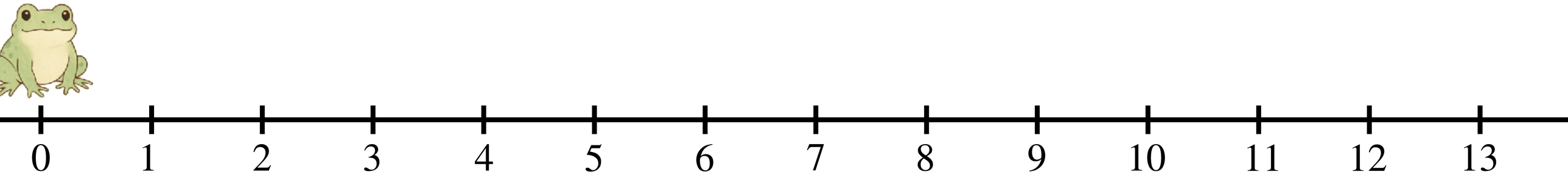
Can AI save the frog?

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Computation and Its Limitations!

Frog picks a Turing Machine M that maps $\mathbb{N} \to \mathbb{N}$.

At time t, frog jumpts to $M(t)$



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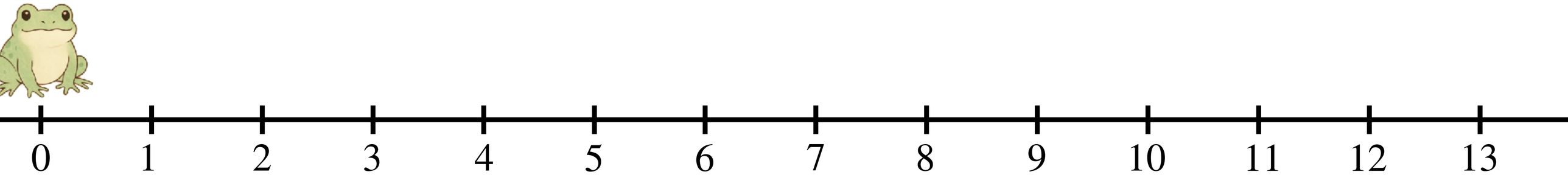

Computation and Its Limitations!

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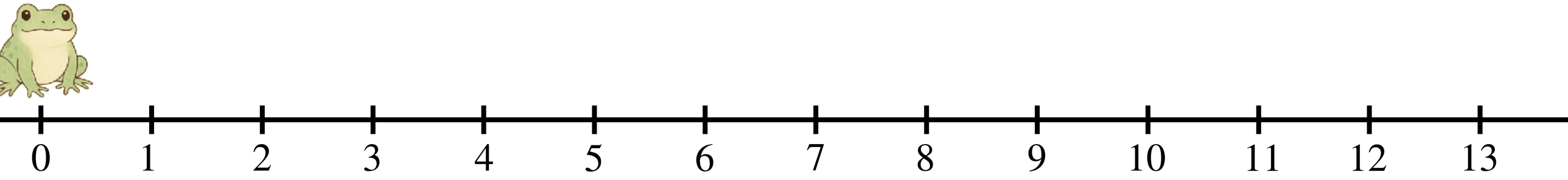
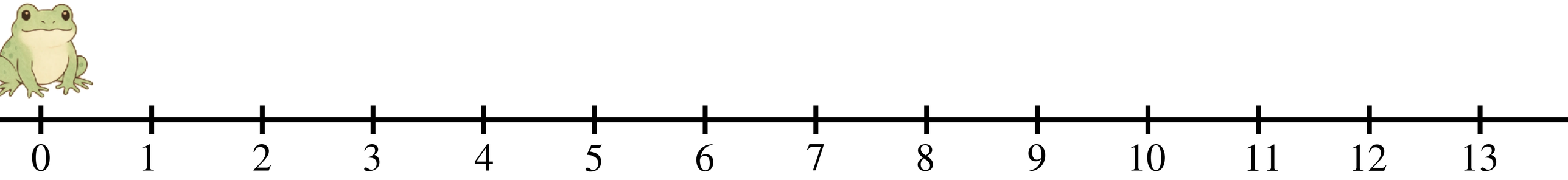
Can the hunter write an
algorithm for their strategy?



The frog can pick from total Turing Machines that map $\mathbb{N} \to \mathbb{N}$ (computable functions). This set is enumerable.



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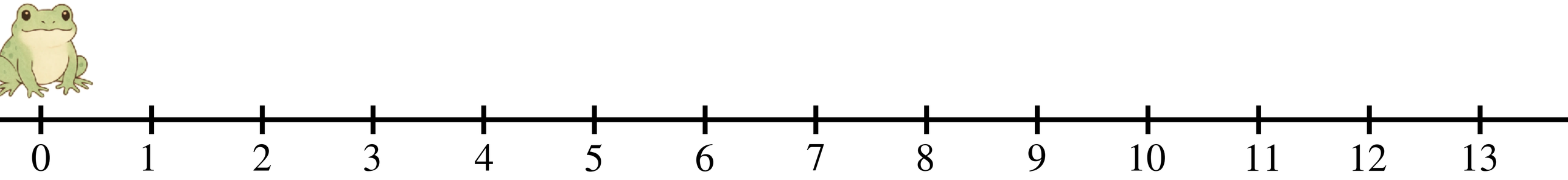
Consider an enumeration $f_1, f_2, \ldots, f_k, \ldots$



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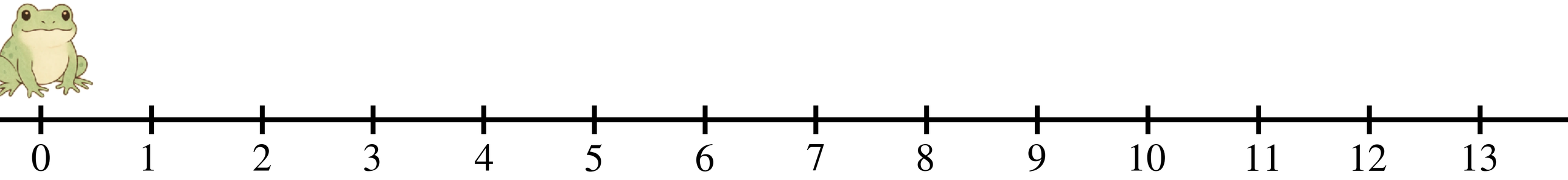

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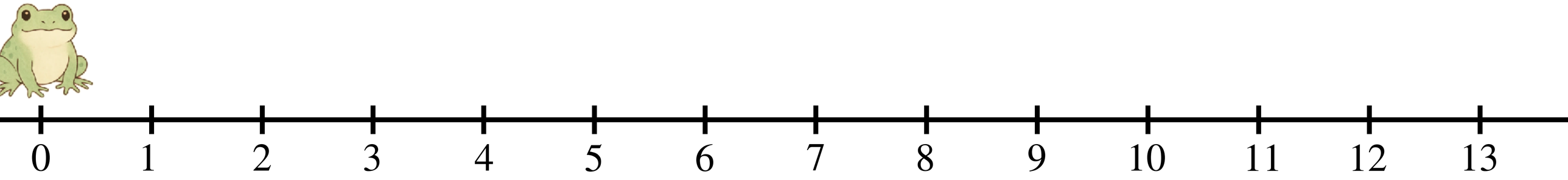
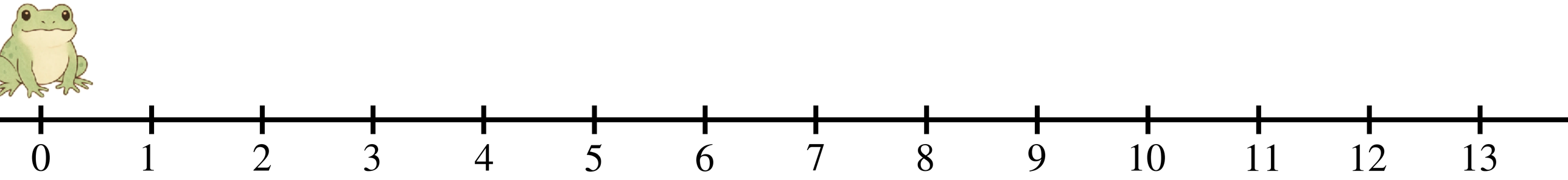
Is $h(t)$ computable?



For sake of contradiction, suppose $h(t)$ was computable.

Then so is $g(t) = h(t) + 1$



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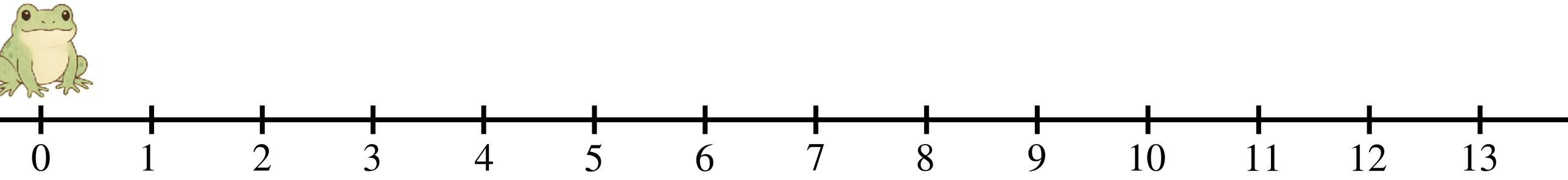

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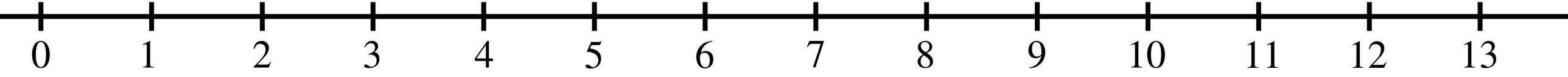
But $g(k) = h(k) + 1 = f_k(k) + 1 \neq f_k(k)$.

$\Rightarrow\!\Leftarrow$



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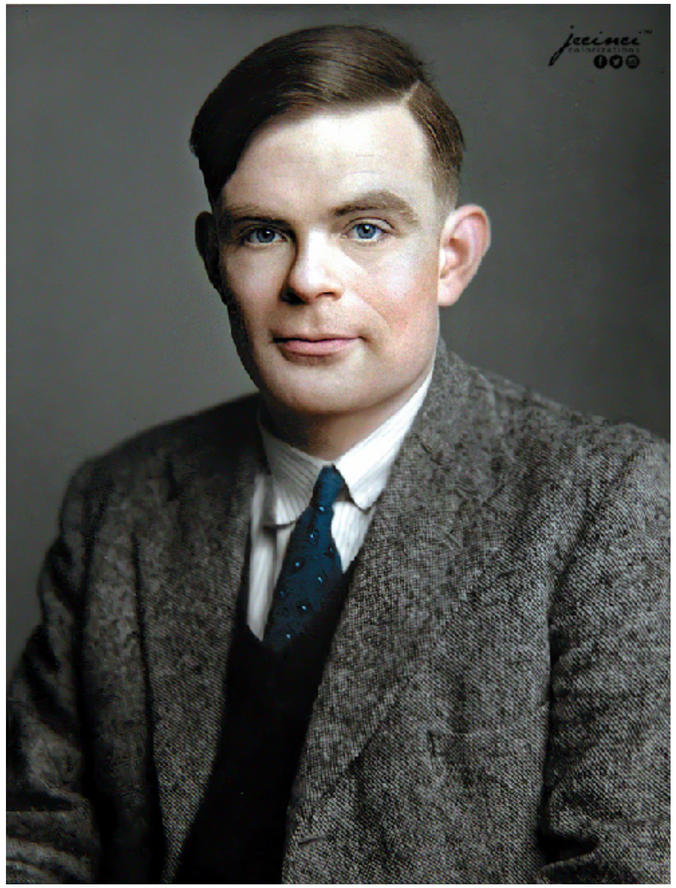

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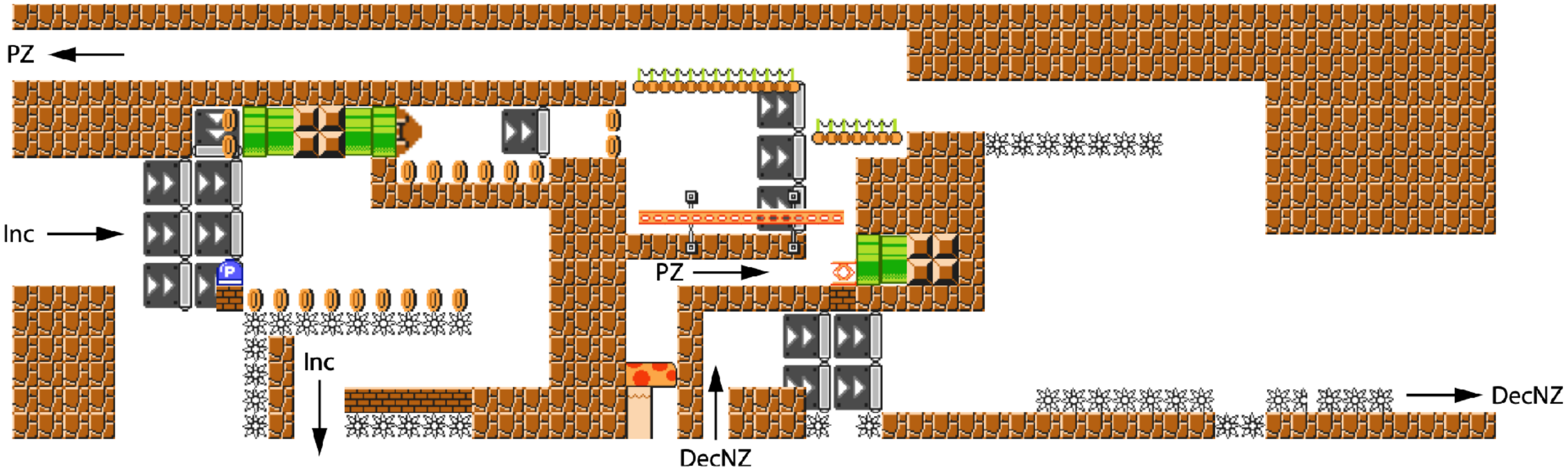
Hunting frogs with computers!



You Can't Solve These Super Mario Bros. Levels: Undecidable Mario Games

MIT Hardness Group* Hayashi Ani† Erik D. Demaine†
Holden Hall† Ricardo Ruiz† Naveen Venkat†

2024.

Tetris and Decidability ⋆

Hendrik Jan Hoogeboom, Walter A. Kosters

Leiden Institute of Advanced Computer Science (LIACS)
Universiteit Leiden, The Netherlands

Abstract

We consider a variant of TETRIS where the sequence of pieces (together with their orientation and horizontal position, which cannot be changed anymore) is generated by a finite state automaton. We show that it is undecidable, given such an automaton, and starting from an empty game board, whether one of the generated sequences leaves an empty game board. This is contrasted with more common situations where piece translations and rotations are allowed.

Key words: Tetris, Decidability, Theory of computation.

1 Introduction

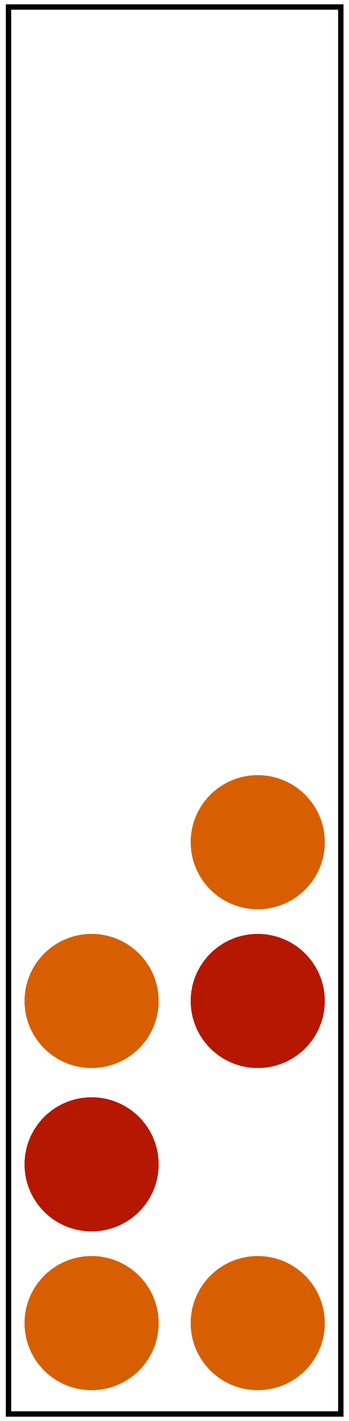
The discovery that the well-known game of TETRIS is provably hard was noticed in the popular press. To be precise, Demaine, Ho-

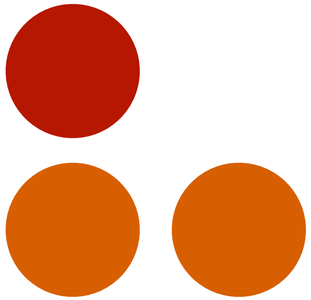
cess. Formulated otherwise, we assume that this process is finite state: the sequence of consecutive pieces together with their orientation and initial position on top of the board is randomly chosen from a given regular language

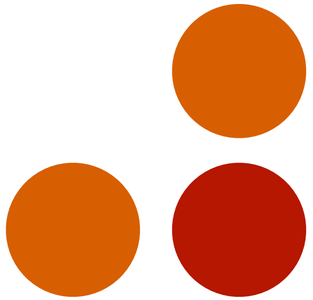
Post Correspondence Problem

Emil Post. *A Variant of a Recursively Unsolvable Problem*. (1946)

Post Correspondence Problem







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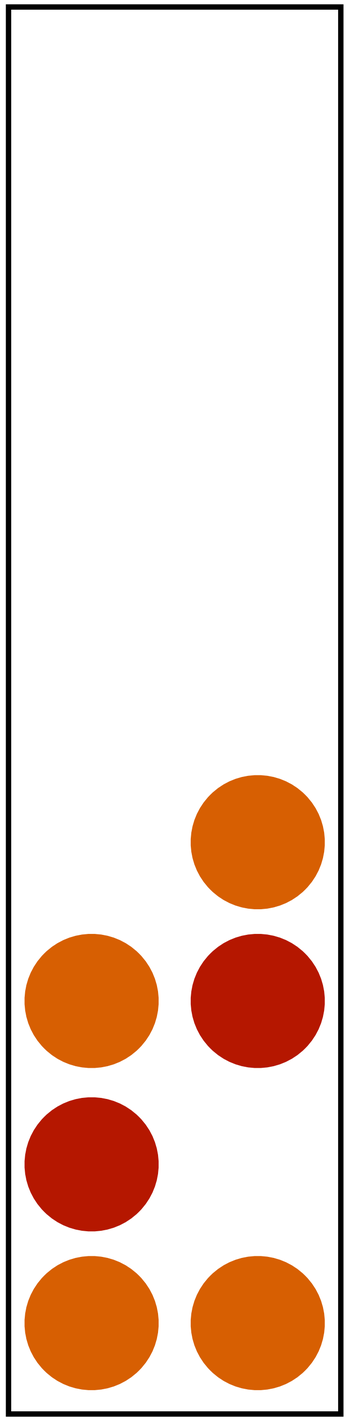

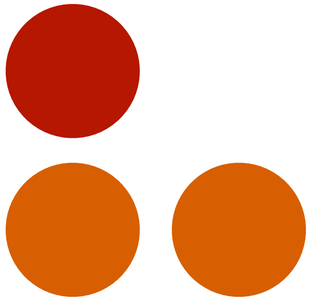

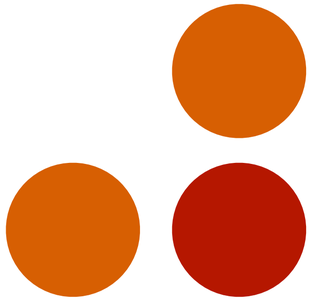

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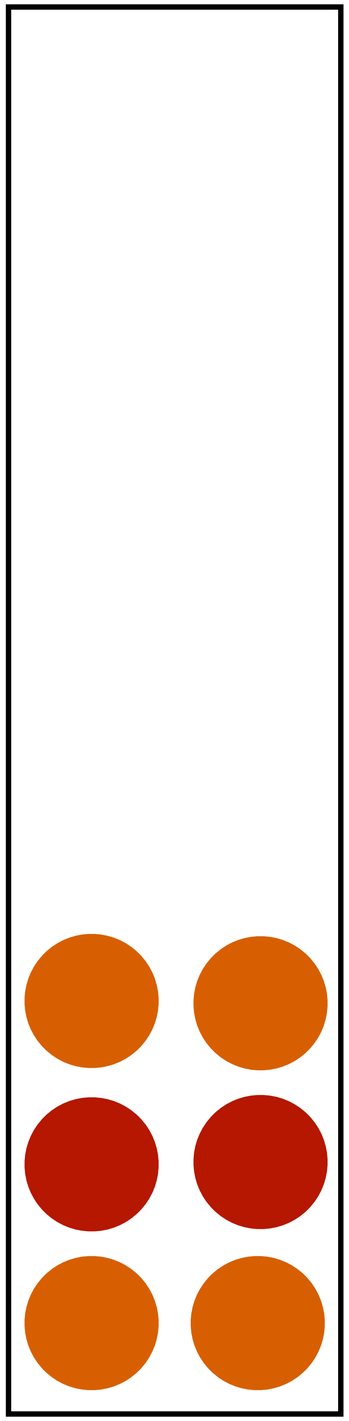

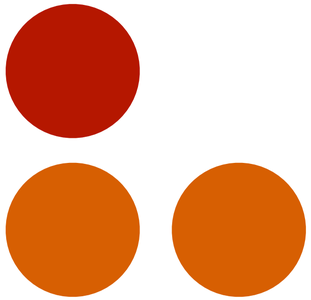

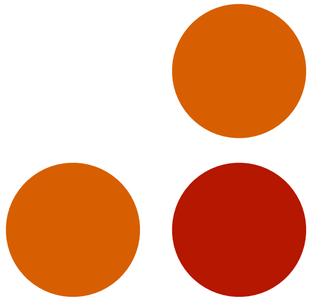

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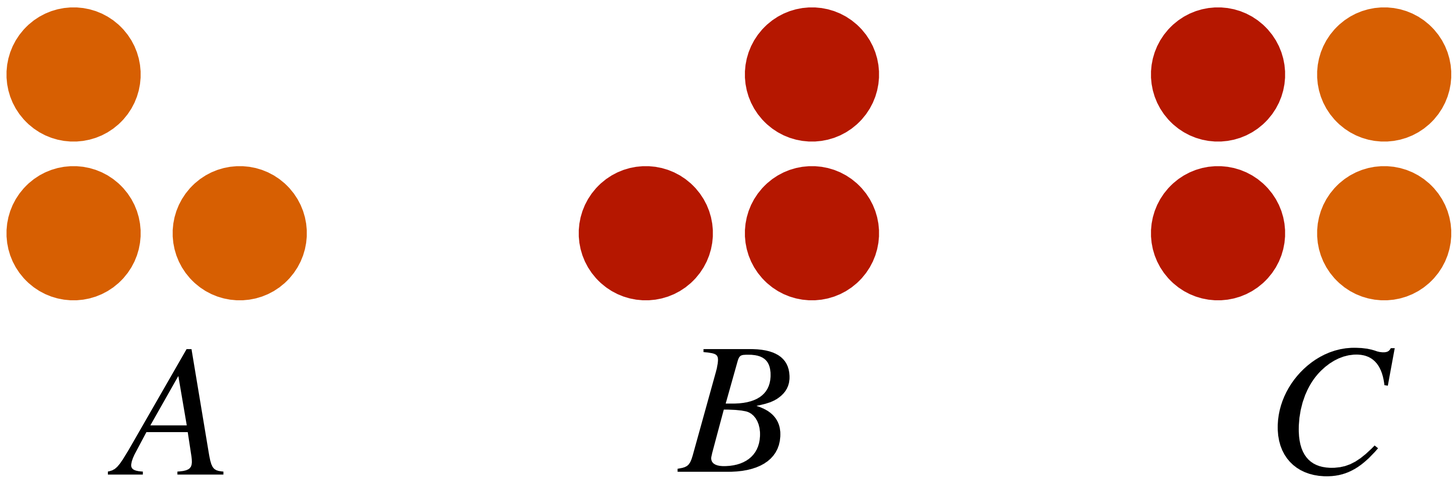
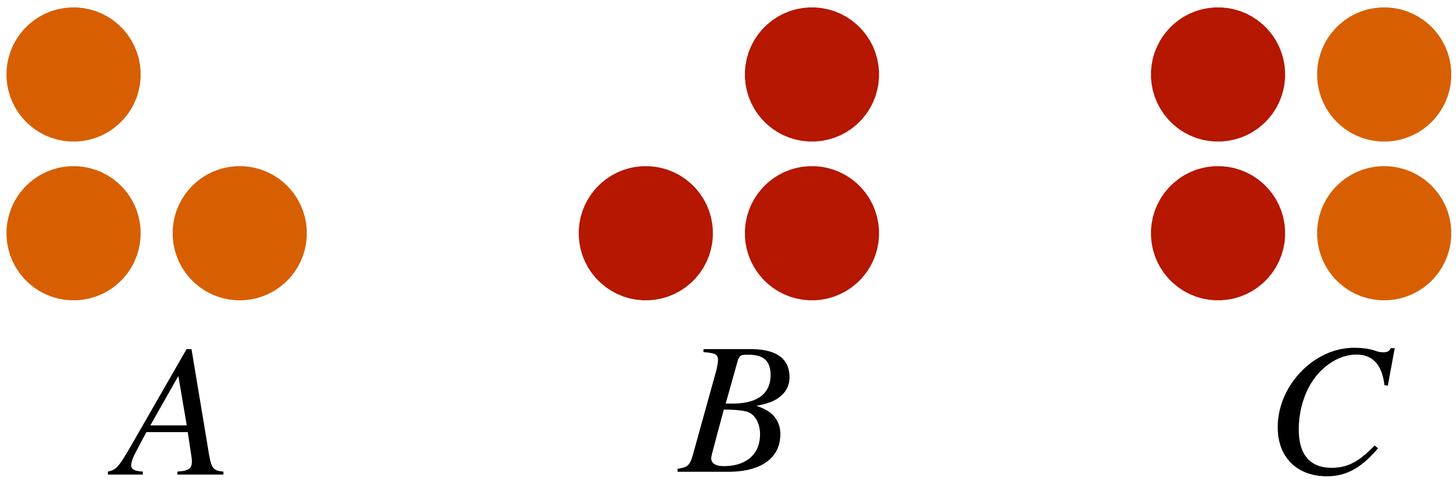


Post Correspondence Problem

$$AACBB$$



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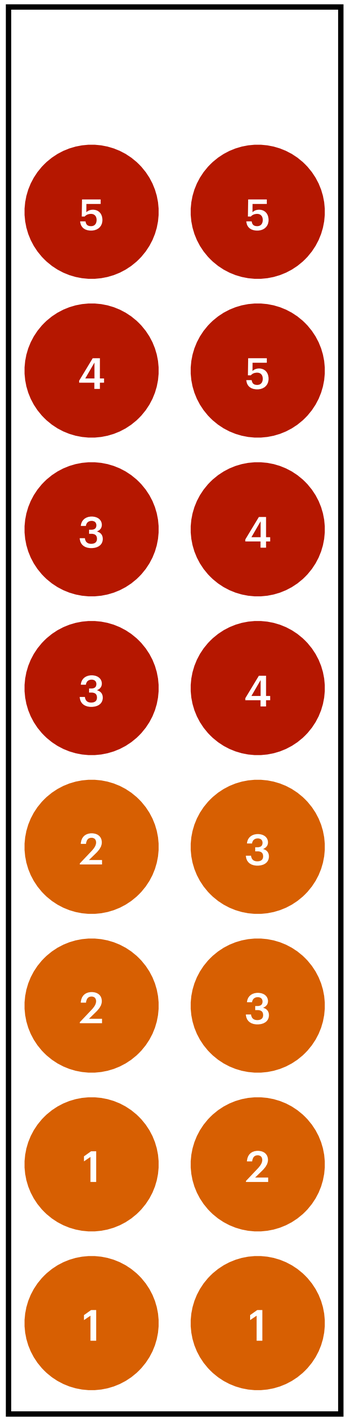

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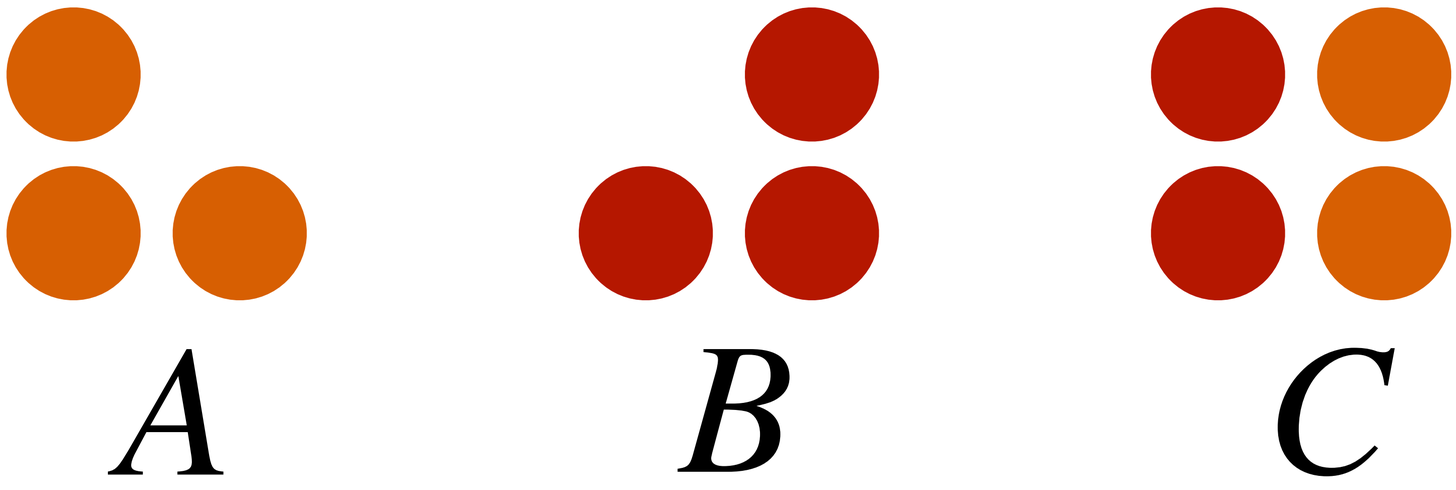

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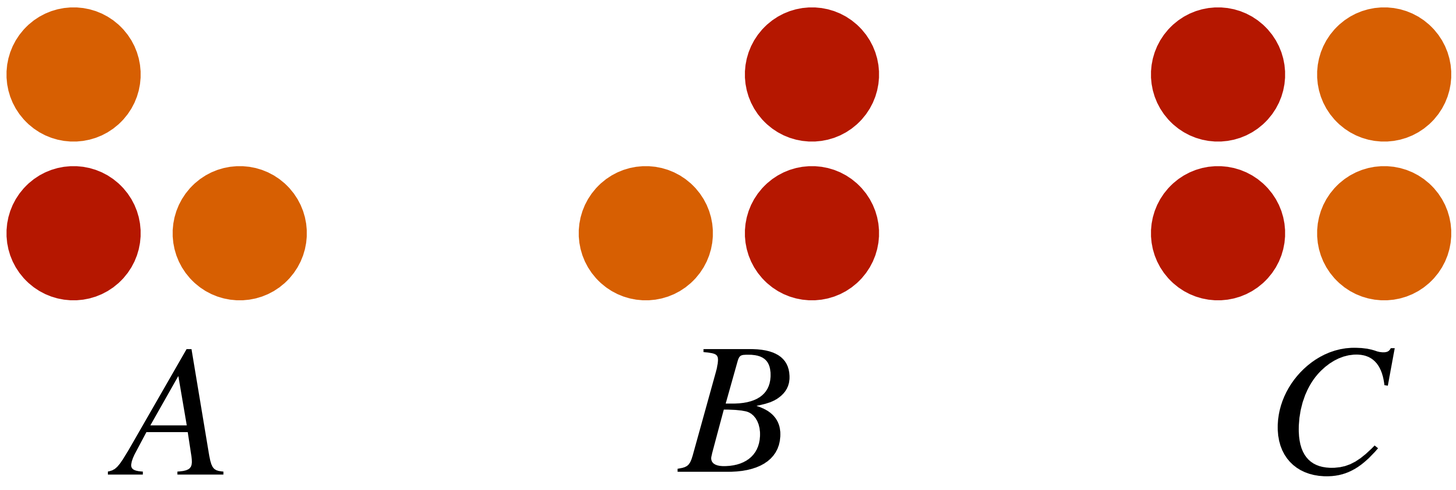

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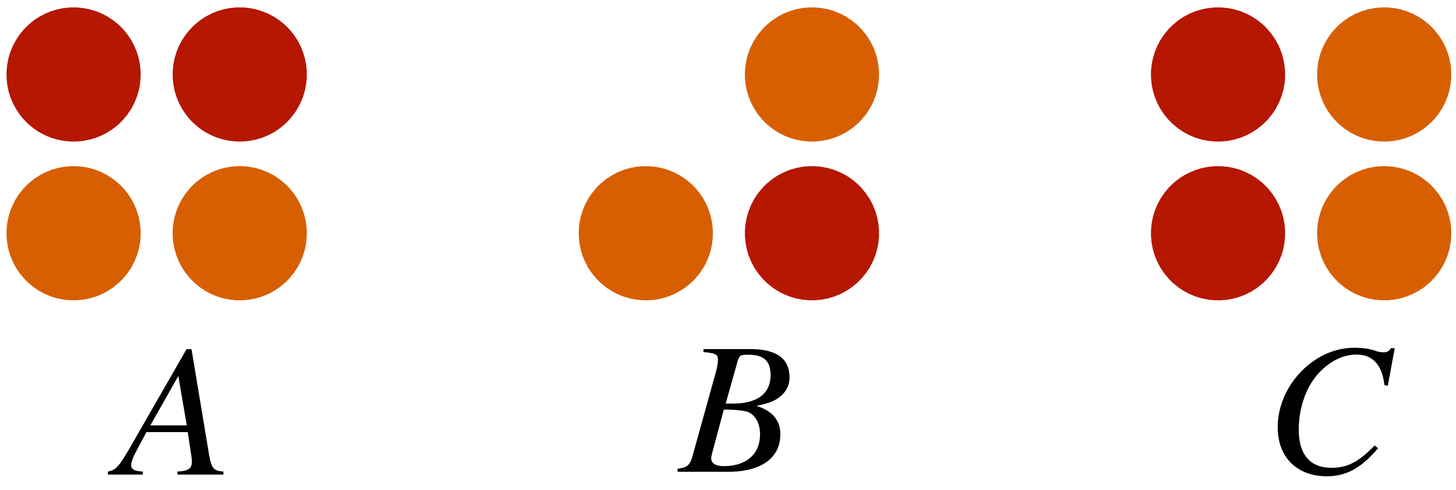
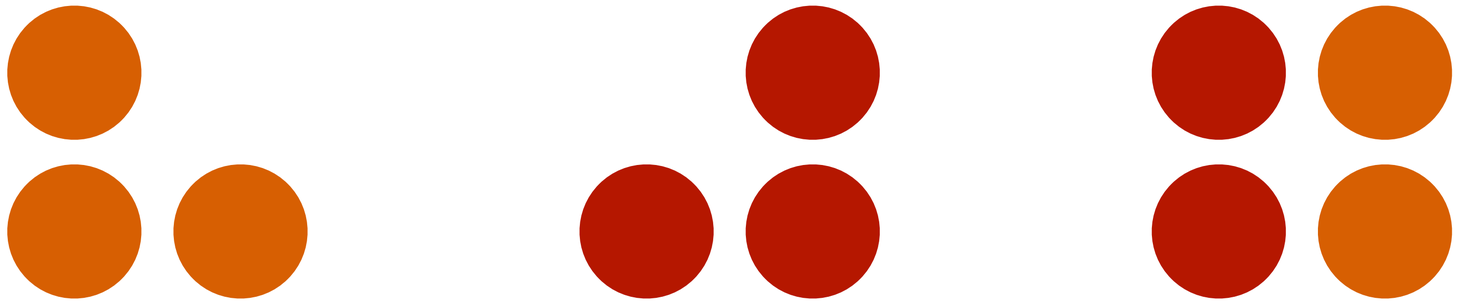


Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \Sigma^*$ and $g : T \to \Sigma^*$, is there a sequence of tiles ($w \in T^*$) such that $\hat{f}(w) = \hat{g}(w)$?



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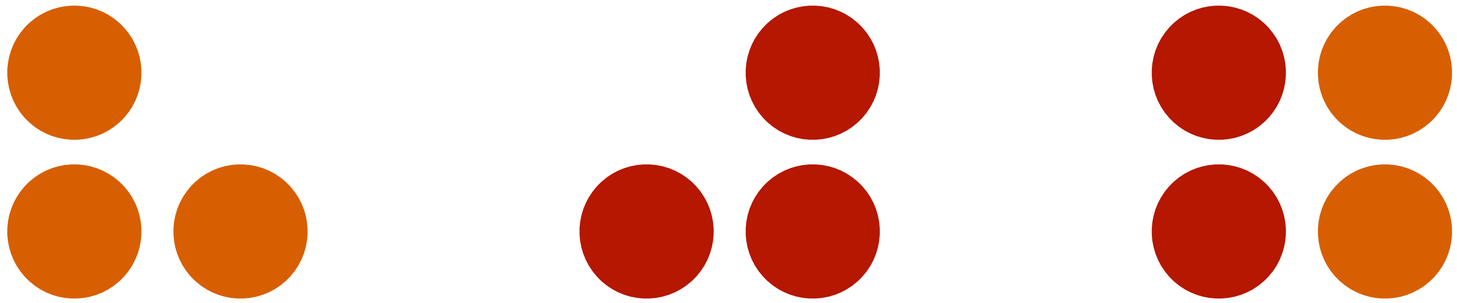

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i	$f(i)$	$g(i)$
1	1	111
2	10111	10
3	10	0

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i	$f(i)$	$g(i)$
1	10	101
2	011	11
3	101	011

Modified Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \Sigma^*$ and $g : T \to \Sigma^*$, is there a $w \in T^*$ such that

$$f(1) \cdot f(w_0) \cdot f(w_1) \cdot \cdots \cdot f(w_n) = g(1) \cdot g(w_0) \cdot g(w_1) \cdot \cdots \cdot g(w_n)?$$

First digit must be 1.

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You prove PCP $\leq_T$ MPCP,

I will prove MPCP $\leq_T$ PCP.

MPCP $\leq_T$ PCP

A TM for PCP can solve MPCP.

i	$f(i)$	$g(i)$
1	1	111
2	10111	10
3	10	0

Suppose I were to solve MPCP on this

i	$f(i)$	$g(i)$
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Suppose I were to solve MPCP on this

i	$f(i)$	$g(i)$
1	1 *	* 1 * 1 * 1
2	1 * 0 * 1 * 1 * 1 *	* 1 * 0
3	1 * 0 *	* 0

MPCP $\leq_T$ PCP

A TM for PCP can solve MPCP.

i	$f(i)$	$g(i)$
1	1	111
2	10111	10
3	10	0

Suppose I were to solve MPCP on this

i	$f(i)$	$g(i)$
0	$*1*$	$*1*1*1$
1	$1*$	$*1*1*1$
2	$1*0*1*1*1*$	$*1*0$
3	$1*0*$	$*0$
4	$\$$	$*\$$

Then I can just solve PCP on this.

$A_{\mathrm{TM}} \leq_T$ **MPCP**

That is given (M, w), there is an instance (T, f, g) of MPCP such that

$$w \in \mathscr{L}(M) \iff (T, f, g) \text{ has a solution}$$

Each partial solution will represent the prefix of an accepting computation history of M on w of the form $\#C_0\#C_1\#\cdots\#C_k\#\cdots$

The MPCP will have a solution when there is an accepting run.

$A_{\mathrm{TM}} \leq_T$ **MPCP**

We will assume that the configuration is represented by the string $\alpha q \beta$ where $\alpha \cdot \beta$ is the tape configuration, the tape marker is at the first letter of β, and q is the control state.

$A_{\mathrm{TM}} \leq_T$ MPCP

We will assume that the configuration is represented by the string $\alpha q \beta$ where $\alpha \cdot \beta$ is the tape configuration, the tape marker is at the first letter of β, and q is the control state.

Design tiles such that $\hat{f}(w)$ lags behind $\hat{g}(w)$.

i	$f(i)$	$g(i)$
1	#	$\#q_0w\#$

$A_{\mathrm{TM}} \leq_T$ **MPCP**

Tape symbols and separator can be copied.

i	$f(i)$	$g(i)$
$\cdots$	X	X
$\cdots$	$\#$	$\#$

$A_{\mathrm{TM}} \leq_T$ **MPCP**

If $\delta(q, X) = (p, Y, R)$

i	$f(i)$	$g(i)$
$\cdots$	qX	Yp

$A_{\mathrm{TM}} \leq_T$ **MPCP**

If $\delta(q, X) = (p, Y, R)$

If $\delta(q, X) = (p, Y, L)$

i	$f(i)$	$g(i)$
$\cdots$	qX	Yp
$\cdots$	ZqX	pZY

$\forall Z \in \Gamma \backslash \{ \vdash \}$

And appropriately for boundary cases…

$A_{\mathrm{TM}} \leq_T$ **MPCP**

Once g reaches the accepting state q...

i	$f(i)$	$g(i)$
$\cdots$	XqY	q
$\cdots$	Xq	q
$\cdots$	qY	q

$\forall X, Y \in \Gamma \backslash \{ \vdash \}$

$A_{\mathrm{TM}} \leq_T$ MPCP

At the end f would have grown to have the same history as g, except that g has an extra $q\#$.

i	$f(i)$	$g(i)$
$\cdots$	$q\#\#$	$\#$

$A_{\mathrm{TM}} \leq_T$ MPCP

At the end f would have grown to have the same history as g, except that g has an extra $q\#$.

i	$f(i)$	$g(i)$
$\cdots$	$q\#\#$	$\#$

Prove that this construction is correct, i.e.,
$w \in \mathscr{L}(M) \iff (T, f, g)$ has a solution.

$A_{\mathrm{TM}} \leq_T$ **MPCP** $\leq_T$ **PCP**

Therefore PCP is undecidable.

The number of tiles depend on the size of the TM.
The number of symbols depend on the tape alphabet.

An Application

Given two CFG G_1 and G_2, it is undecidable if $\mathscr{L}(G_1) \cap \mathscr{L}(G_2) = \emptyset$.

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$$S \to (f(i))^R S g(i) \;\mid\; (f(1))^R \$ g(1) \qquad \forall i \in T$$

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Let G_2 be the grammar of palindromes.

An Application

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Suppose not.

$\mathscr{L}(G_1) \cap \mathscr{L}(G_2) = \varnothing \iff (T, f, g)$ does not have a solution

Therefore, intersection emptiness of CFGs is undecidable.

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Suppose not.

$\mathscr{L}(G_1) \cap \mathscr{L}(G_2) = \varnothing \iff (T, f, g)$ does not have a solution

Therefore, intersection emptiness of CFGs is undecidable.

Another proof of CFG are not closed under intersection!

Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \Sigma^*$ and $g : T \to \Sigma^*$, is there a sequence of tiles ($w \in T^+$) such that $\hat{f}(w) = \hat{g}(w)$?

Suppose $\Sigma = \{a\}$. Prove that PCP is decidable.

Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \mathbb{N}$ and $g : T \to \mathbb{N}$, is there a $w \in T^+$ such that $f(w_0) + f(w_1) + \cdots + f(w_n) = g(w_0) + g(w_1) + \cdots + g(w_n)$?

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If for some $i \in T$, $d(i)$ is zero then we are done.

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Let $d(i) = f(i) - g(i)$

If for some $i \in T$, $d(i)$ is zero then we are done.

If for all $i \in T$, $d(i)$ is positive then there is no solution.

If for all $i \in T$, $d(i)$ is negative then there is no solution.

Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \mathbb{N}$ and $g : T \to \mathbb{N}$, is there a $w \in T^+$ such that $f(w_0) + f(w_1) + \cdots + f(w_n) = g(w_0) + g(w_1) + \cdots + g(w_n)$?

Let $d(i) = f(i) - g(i)$

For some $i_0, i_i \in T$, $d(i_0) < 0 < d(i_1)$.

Let $n = -d(i_0)$ and $m = d(i_1)$.

Post Correspondence Problem

Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \mathbb{N}$ and $g : T \to \mathbb{N}$, is there a $w \in T^+$ such that $f(w_0) + f(w_1) + \cdots + f(w_n) = g(w_0) + g(w_1) + \cdots + g(w_n)$?

Let $d(i) = f(i) - g(i)$

For some $i_0, i_i \in T$, $d(i_0) < 0 < d(i_1)$.

Let $n = -d(i_0)$ and $m = d(i_1)$.

Consider $w = i_0^m i_1^n$.

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Given a fixed set of tiles $T = \{1, \ldots, k\}$, and functions $f : T \to \Sigma^*$ and $g : T \to \Sigma^*$, is there a sequence of tiles ($w \in T^+$) such that $\hat{f}(w) = \hat{g}(w)$?

Suppose $|\Sigma| \geq 2$. Now what?

If $|T| = 1$ it is decidable.

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If $|T| = 1$ it is decidable.

If $|T| = 2$ it is decidable.

Ehrenfeucht, Karhumäki, Rozenberg. *The (generalized) post correspondence problem with lists consisting of two words is decidable*. (1982)

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Suppose $|\Sigma| \geq 2$. Now what?

If $|T| = 1$ it is decidable.

If $|T| = 2$ it is decidable.

If $|T| \geq 5$ it is undecidable.

Turlough Neary. *Undecidability in Binary Tag Systems and the Post Correspondence Problem for Five Pairs of Words*. (2015).

Post Correspondence Problem

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Suppose $|\Sigma| \geq 2$. Now what?

If $|T| = 1$ it is decidable.

If $|T| = 2$ it is decidable.

$|T| = 3$ or $|T| = 4$ is unknown!

If $|T| \geq 5$ it is undecidable.